Worked Answer Key: Practice Test 1.1–1.6

1. (1.1) Table: $f = 8, 3, 0, -1, 0, 3$ at $x = -2, \ldots, 3$.

- First differences (each step of 1): $3 - 8 = -5$, $0 - 3 = -3$, $-1 - 0 = -1$, $0 - (-1) = 1$, $3 - 0 = 3$.
- (a) Negative differences until $x = 1$, then positive. **Decreasing on $(-2, 1)$, increasing on $(1, 3)$.** The lowest point in the table is $(1, -1)$.
- (b) Zeros are where $f(x) = 0$: $\boxed{x = 0 \text{ and } x = 2}$.
- (c) The differences $-5, -3, -1, 1, 3$ are increasing (the rate of change is increasing), so f is **concave up**. Bonus: the second differences are all 2, so it's quadratic: $f(x) = x^2 - 2x$.

2. (1.2) $f(x) = x^3$ on $[1, 3]$.

- $f(3) = 27$ and $f(1) = 1$.
- $\dfrac{27 - 1}{3 - 1} = \dfrac{26}{2} = \boxed{13}$

3. (1.2) 50°F at 2 PM, 62°F at 6 PM.

- $\dfrac{62 - 50}{6 - 2} = \dfrac{12}{4} = \boxed{3 \text{ °F per hour}}$
- Meaning: between 2 and 6 PM the temperature rose an **average** of 3 degrees each hour. It may not have risen exactly 3 degrees in every single hour.

4. (1.2) Rate of change of $g(x) = x^2 + 1$ at $x = 2$.

- $g(2) = 5$, $g(2.1) = 4.41 + 1 = 5.41$, $g(1.9) = 3.61 + 1 = 4.61$.
- Right side: $\dfrac{5.41 - 5}{2.1 - 2} = \dfrac{0.41}{0.1} = 4.1$
- Left side: $\dfrac{5 - 4.61}{2 - 1.9} = \dfrac{0.39}{0.1} = 3.9$
- The true rate is between them: $\boxed{\approx 4}$. (Smaller intervals, like $[2, 2.01]$, would close in further.)

5. (1.3) Linear, quadratic or neither?

- (a) $1, 4, 7, 10, 13$: first differences $3, 3, 3, 3$, constant. **Linear.**
- (b) $3, 4, 7, 12, 19$: first differences $1, 3, 5, 7$; second differences $2, 2, 2$, constant. **Quadratic.**
- (c) $1, 2, 4, 8, 16$: first differences $1, 2, 4, 8$; second differences $1, 2, 4$, not constant. **Neither.** (The outputs double each time: constant ratio 2, so it's exponential.)

6. (1.3) $f(x) = 2x^2$.

- Values: $f(0) = 0$, $f(1) = 2$, $f(2) = 8$, $f(3) = 18$.
- Rates: $[0, 1]$: $\frac{2-0}{1} = 2$. $[1, 2]$: $\frac{8-2}{1} = 6$. $[2, 3]$: $\frac{18-8}{1} = 10$.
- They go up by **4** each time. For a quadratic the average rate of change changes at a constant rate. That's the 1.3 fact being tested.

7. (1.4) $f(x) = (x-1)(x-2)(x-3)$.

- (a) Zeros: $\boxed{1, 2, 3}$.
- (b) Test a point in each interval:
  - $(1, 2)$, test $x = 1.5$: $(0.5)(-0.5)(-1.5) = +0.375$, so **positive**.
  - $(2, 3)$, test $x = 2.5$: $(1.5)(0.5)(-0.5) = -0.375$, so **negative**.
- (c) Degree 3, so at most $3 - 1 = \boxed{2}$ turning points.

8. (1.4) Degree 5.

- At most **5** real zeros (the degree).
- At most **4** turning points (degree − 1).
- At least **1** real zero. Odd degree means the two ends go in opposite directions (one up, one down), so the graph has to cross the x-axis at least once. (Also, complex zeros come in pairs, so 5 zeros can't all be complex.)

9. (1.5) Zeros of $x^4 - 16$.

- Difference of squares: $(x^2)^2 - 4^2 = (x^2 - 4)(x^2 + 4)$.
- Again: $x^2 - 4 = (x - 2)(x + 2)$.
- $x^2 + 4 = 0$ gives $x^2 = -4$, so $x = \pm 2i$.
- Zeros: $\boxed{2,\ -2,\ 2i,\ -2i}$ (4 zeros for degree 4 ✓).

10. (1.5) $x^2 + 2x + 5 = 0$.

- Doesn't factor, so use the quadratic formula with $a = 1$, $b = 2$, $c = 5$.
- Discriminant: $b^2 - 4ac = 4 - 20 = -16$. Negative means complex zeros.
- $x = \dfrac{-2 \pm \sqrt{-16}}{2} = \dfrac{-2 \pm 4i}{2} = \boxed{-1 \pm 2i}$

11. (1.5) Least degree, zero 3 (multiplicity 2) and zero i.

- Real coefficients mean $-i$ must also be a zero. Zeros: $3, 3, i, -i$, so degree 4.
- $(x - 3)^2 = x^2 - 6x + 9$ and $(x - i)(x + i) = x^2 - i^2 = x^2 + 1$.
- Multiply: $(x^2 - 6x + 9)(x^2 + 1) = x^4 + x^2 - 6x^3 - 6x + 9x^2 + 9$.
- Combine: $\boxed{x^4 - 6x^3 + 10x^2 - 6x + 9}$

12. (1.5) $f(x) = x^2(x - 4)^3$.

- $x = 0$: factor x^2, multiplicity **2** (even), so the graph **bounces**.
- $x = 4$: factor $(x - 4)^3$, multiplicity **3** (odd), so it **crosses**, flattening out as it passes through (like x^3 at the origin).

13. (1.5) Plug in $-x$ and compare.

- (a) $(-x)^5 - 2(-x)^3 = -x^5 + 2x^3 = -(x^5 - 2x^3) = -f(x)$, so **odd**.
- (b) $(-x)^4 + (-x)^2 - 7 = x^4 + x^2 - 7 = f(x)$, so **even**.
- (c) $(-x)^2 + (-x) = x^2 - x$. That's not $f(x) = x^2 + x$, and not $-f(x) = -x^2 - x$, so **neither**.
- Shortcut: all even powers (a constant counts as x^0) means even, and all odd powers means odd. A mix means neither.

14. (1.6) End behavior.

- (a) Reorder: $-x^7 + 4x^6 + 1$. The leading term is $-x^7$: odd degree, negative. $\lim_{x \to -\infty} f(x) = \infty$ and $\lim_{x \to \infty} f(x) = -\infty$.
- (b) Leading term $-(x^2)(x^2) = -x^4$: even degree, negative. $\lim_{x \to \pm\infty} g(x) = -\infty$ (down on both ends).
- (c) Leading term $2x \cdot x \cdot x^2 = 2x^4$: even degree, positive. $\lim_{x \to \pm\infty} h(x) = \infty$ (up on both ends).

15. (1.4–1.6) Sketch $f(x) = (x+1)^2(x-2)$.

- Zeros: $x = -1$ (multiplicity 2, **bounce**) and $x = 2$ (multiplicity 1, **cross**).
- y-intercept: $f(0) = (1)^2(-2) = -2$, so $(0, -2)$.
- Leading term $x^2 \cdot x = x^3$: degree 3, positive, so down on the left and up on the right.
- Putting it together: start at the bottom left, rise up to touch the x-axis at -1 and turn back down, pass through $(0, -2)$, turn around (somewhere between 0 and 2), cross up through $x = 2$, and keep rising.

16. (AP-style) Water level $W(t)$: $20, 25, 36, 47, 52, 45$ at $t = 0, \ldots, 5$.

- (a) $\dfrac{W(4) - W(1)}{4 - 1} = \dfrac{52 - 25}{3} = \boxed{9 \text{ inches per hour}}$. Between hour 1 and hour 4, the water level rose an average of 9 inches each hour.
- (b) One-hour changes: $+5, +11, +11, +5, -7$. The biggest rise is **11 in/hr**, on $[1, 2]$ and $[2, 3]$.
- (c) On $[3, 5]$ the changes are $+5$ then -7. The rate of change is **decreasing**, so W is **concave down** there.
- (d) The level rises until $t = 4$ (change $+5$ on $[3, 4]$) and then falls (change -7 on $[4, 5]$). It switches from increasing to decreasing **at about** $t = 4$, so that's the local maximum (about 52 inches).
- AP grading tip: each part needs the number, the units, and a reason that cites the table.